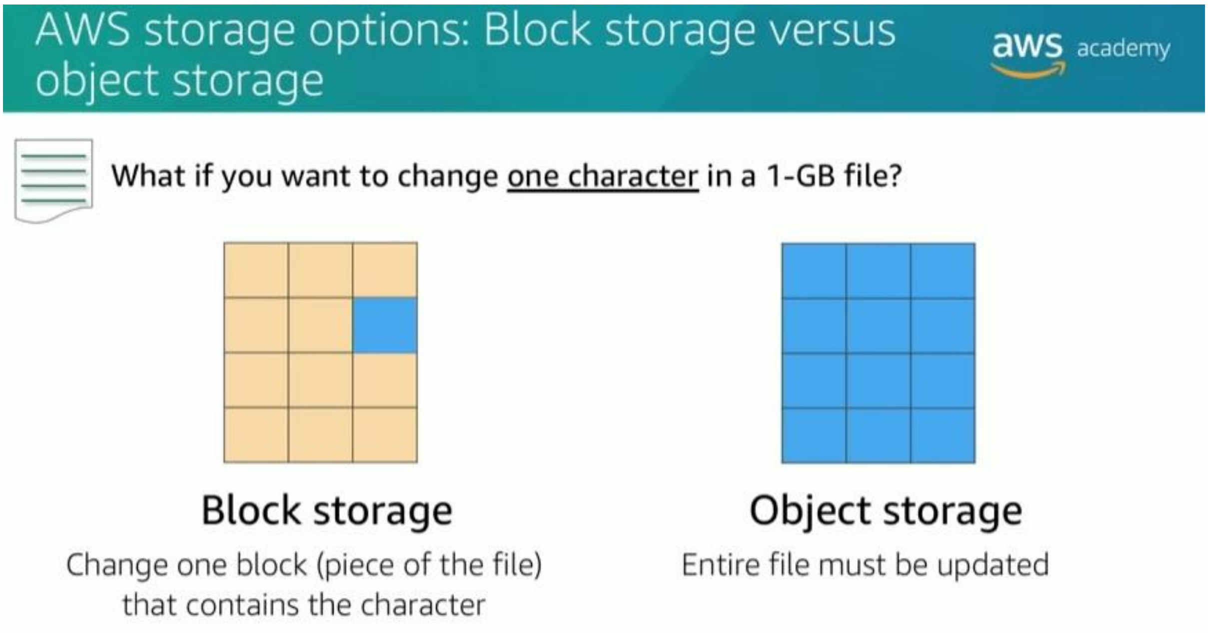
Module 7: Storage

1. AWS EBS



Amazon EBS

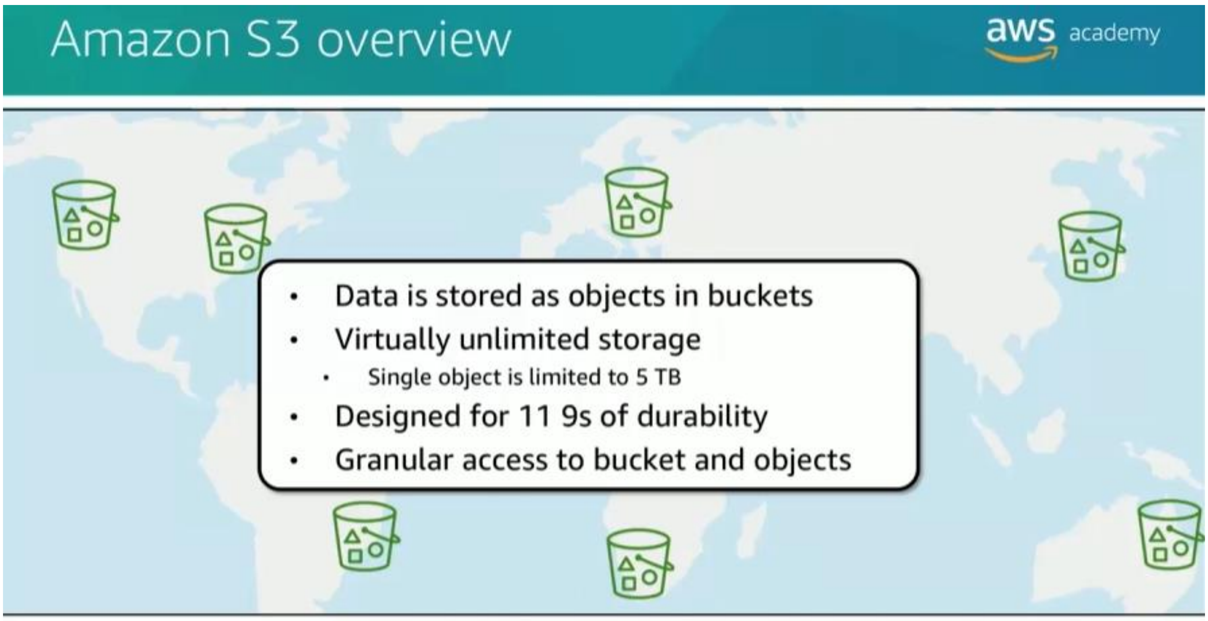
Amazon EBS enables you to **create individual storage volumes** and **attach them** to an Amazon EC2 instance:

- Amazon EBS offers block-level storage.
- Volumes are automatically replicated within its Availability Zone.
- It can be backed up automatically to Amazon S3 through snapshots.
- Uses include –
  - Boot volumes and storage for Amazon Elastic Compute Cloud (Amazon EC2) instances
  - Data storage with a file system
  - Database hosts
  - Enterprise applications

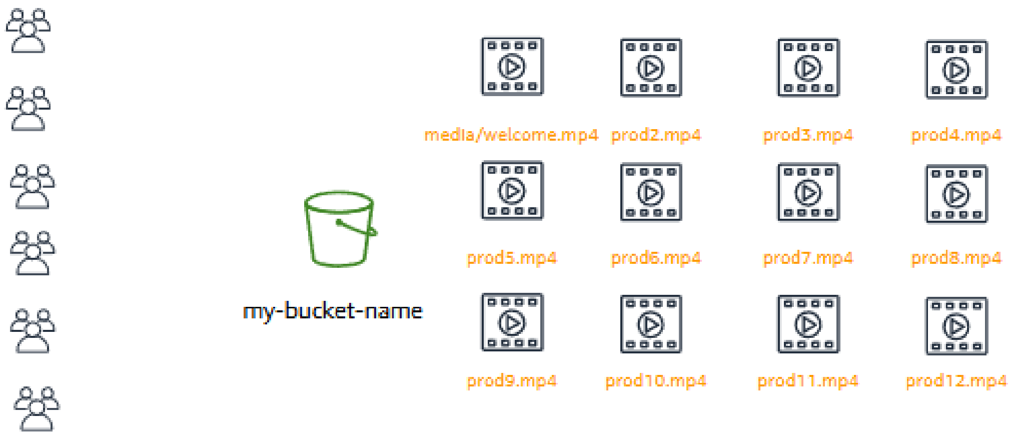
Amazon EBS features

- Snapshots –
  - Point-in-time snapshots
  - Recreate a new volume at any time
- Encryption –
  - Encrypted Amazon EBS volumes
  - No additional cost
- Elasticity –
  - Increase capacity
  - Change to different types

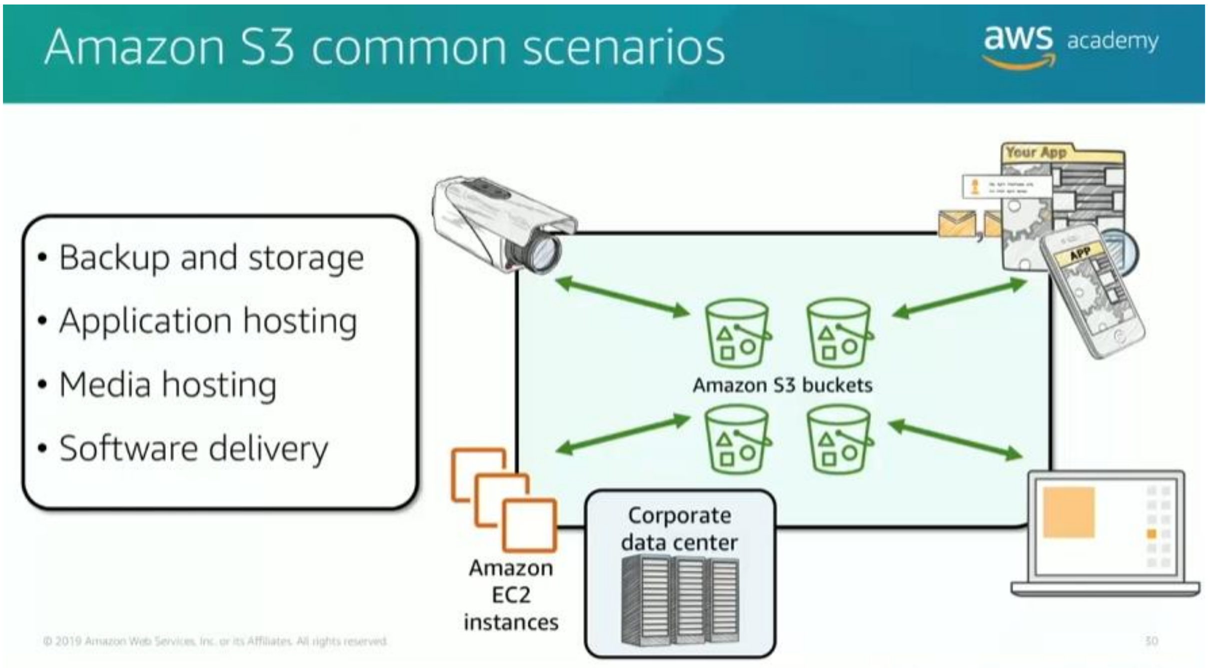
2. AWS S3



Designed for seamless scaling

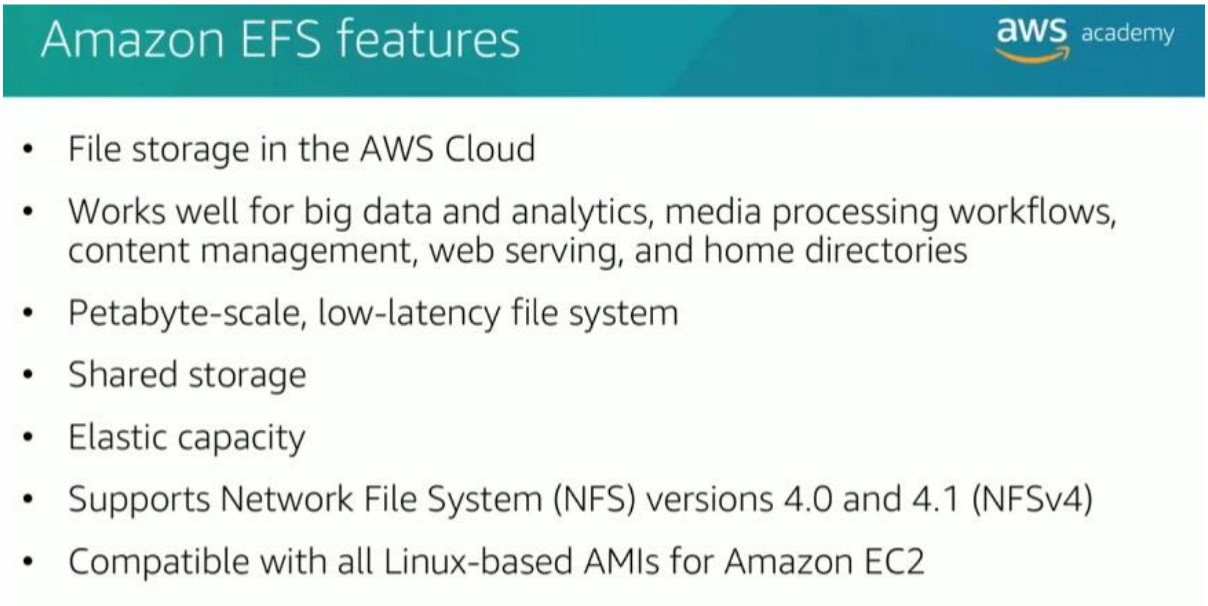


Access the data anywhere



3. AWS EFS

Amazon Elastic File System (Amazon EFS) provides simple, scalable, elastic file storage for use with AWS services and on-premises resources. It offers a simple interface that enables you to create and configure file systems quickly and easily.



Amazon EFS implementation

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1. Create your Amazon EC2 resources and launch your Amazon EC2 instance.
2. Create your Amazon EFS file system.
3. Create your mount targets in the appropriate subnets.
4. Connect your Amazon EC2 instances to the mount targets.
5. Verify the resources and protection of your AWS account.

4. AWS S3 Glacier

Amazon S3 Glacier review

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Amazon S3 Glacier is a **data archiving service** that is designed for **security**, **durability**, and an **extremely low cost**.

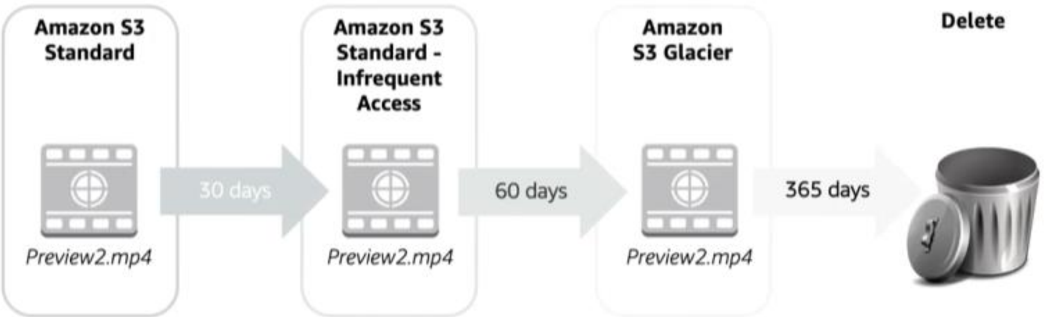
- Amazon S3 Glacier is designed to provide 11 9s of durability for objects.
- It supports the encryption of data in transit and at rest through Secure Sockets Layer (SSL) or Transport Layer Security (TLS).
- The Vault Lock feature enforces compliance through a policy.
- Extremely low-cost design works well for long-term archiving.
  - Provides three options for access to archives—expedited, standard, and bulk—retrieval times range from a few minutes to several hours.

Amazon S3 Glacier use cases

- Media asset archiving
- Healthcare information archiving
- Regulatory and compliance archiving
- Scientific data archiving
- Digital preservation
- Magnetic tape replacement

Lifecycle policies

Amazon S3 lifecycle policies enable you to delete or move objects based on age.



Security with Amazon S3 Glacier

Amazon S3 Glacier

Control access with IAM

Amazon S3 Glacier encrypts your data with AES-256

Amazon S3 Glacier manages your keys for you